

KRZYSZTOF SZUMKO

SENIOR SOFTWARE ENGINEER

Somerset, UK (Remote) | www.linkedin.com/in/kszumko | ILR (UK) & EU Citizen

Profile Summary

8+ years building high-stakes production platforms in healthcare and IoT, specialising in backend Python, distributed systems, and safety-critical architecture. Scaled systems to 3K requests/second, cut infrastructure costs by up to 80%, and delivered EU GMP Annex 11-compliant platforms with legally mandated audit trails. Proven track record leading cross-functional teams and owning architecture end-to-end in environments where failure carries real-world consequences.

Technical Skills

Languages: Python, Rust, JavaScript, Java

Python & Backend: Django, Django REST Framework, FastAPI, Flask, Django-Ninja, REST APIs, OpenAPI/Swagger, OAuth 2.0, WebSockets

Infrastructure & Automation: Linux, AWS, Docker, Nginx, Terraform, CloudFormation, Apache Airflow, GitHub Actions, CI/CD

Distributed Systems: Celery, RabbitMQ, Redis, Microservices, Event-driven Architecture, Distributed Task Processing

Databases & Data: PostgreSQL, MySQL, Trino/Starburst, Data Lake Architectures

Observability: Dynatrace, Datadog, AWS CloudWatch, Metrics, Alerting, Dashboards

Frontend: Vue 3, React

Testing & Quality: pytest, Django Test, Playwright, vue-jest, TDD, Ruff, API Contract Testing

Professional Experience

Senior Software Engineer — Freelance

Sigma Code | Aug 2024 – Present

(client Quantum Cryogenics, Full-time from Nov 2025)

Tech: Python · Django · Vue 3 · AWS · REST APIs · Celery · PostgreSQL · RabbitMQ

- **Modernised Django** – upgraded from Django 3.2 LTS to Django 5.2 LTS, addressing breaking changes, deprecated functionality, and dependency compatibility
- **Owned CryoHub Cloud V2 end-to-end** – Django + Vue 3 platform for cryogenic telemetry, alarms, and mobile/web monitoring (3.5k+ commits)
- **Reduced critical API latency 100×** (30s to 300ms) via query optimisation, caching, and performance test suites
- **Built sub-second real-time alerting** with fan-out delivery, severity routing, and audit trails (Celery, WebSockets, Novu)
- **Re-architected alarm engine for V2** – configurable rules, cooldown policies, auto-clearance jobs, and admin tooling replacing hardcoded Gen 1 logic
- **Delivered CSV/PDF reporting and analytics suite** – heatmaps, gap analysis, and synchronised telemetry graphs
- **Engineered resilient Celery/RabbitMQ cluster** infrastructure with dedicated queues, retry policies, and AWS CloudWatch heartbeats
- **Hardened AWS/IONOS production systems** – secrets management, TLS, CORS, lockouts, and CI pipelines

Senior Software Engineer – Acting Team Lead

OPTUM UK (EMIS) | July 2024 – November 2025

Tech: Python (Flask, FastAPI) · Airflow · Celery · Redis · PostgreSQL · Starburst/Trino · Apache Superset · AWS (CloudFormation, Terraform) · Docker · Dynatrace · Datadog · React · Java · Microservices

Data Out Team (Jul 2024 – Jan 2025)

- Maintained and developed the **core data lake platform** — Starburst/Trino cluster, auth-routing proxy, and Apache Superset (customer-facing SQL interface)
- **Rebuilt the data lake auth proxy** to handle 3K requests/second (up from 1K) at the same cost, delivering 3x throughput improvement and slashing latency for time-sensitive clinical queries

Acting Team Lead — Data Management (Jan 2025 – Nov 2025)

- **Improved sprint throughput by ~60%** — Led 9 engineers in an Agile environment — sprint planning, quarterly goals, and stakeholder demos
- **Drove 44–66% reduction in Airflow cluster costs** through right-sizing hardware to 80th percentile peak usage, off-hours environment switching, and introducing smarter auto-scaling
- **Led Airflow v2 upgrade and prototyped v3 migration**
- **Improved production observability** — steered Data Management's migration from Datadog to Dynatrace, consolidating dashboards and introducing alerts for critical platform metrics.

Senior Lead Python Developer

Quantum Cryogenics | August 2019 – July 2024

Tech: Python (Django, DRF, Django-Ninja) · Celery · RabbitMQ · Redis · PostgreSQL · MySQL · AWS · Vue 3 · Rust (Tauri) · Raspberry Pi/LattePanda (SBC) · eCharts

Architected and delivered two safety-critical cryogenic platforms (CryoHub Cloud, Polaris Vector) under EU GMP Annex 11 compliance obligations — mandatory 4-hour monitoring continuity with legally auditable trails for universities and enterprise clients.

- **Architected CryoHub Cloud from greenfield** — data model, APIs, compliance workflows, and reporting — through to production rollout
- **Achieved EU GMP Annex 11 compliance** through tamper-evident audit trails, attributable user actions, controlled workflows, and role-based access controls for all safety-critical operations
- Built multi-tier alarm propagation and escalation logic across both platforms, covering generation, acknowledgement, muting, and feedback states with full auditability
- **Designed safety-critical hazardous gas monitoring** (O₂, CO₂) for occupied buildings, where suppressed or missed alerts trigger mandatory evacuation procedures
- **Delivered Polaris Vector IoT platform** — led an 8-person cross-functional team to production rollout in 8 months, supporting over 100 connected devices.
- **Delivered traceable ATEX device management** — built a Django API integration with an external production data warehouse supporting 10,000+ devices.
- **Reduced infrastructure spend by approximately 80%** through AWS architecture redesign
- **Delivered 10x memory optimisation** through Rust-based embedded development (Tauri) on resource constrained hardware
- **Managed a 3-person software team**, set code quality and security standards, and provided tier-3 support across all Quantum products

Education

Open University UK | 2016 – 2020

BSc (Honours) Computing & IT (Software Engineering)

First-Class Distinction in Java Development (60-credit year-long module)